Neurocomputing 685 (2026) 133641

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

Self-supervised saliency-guided dual-attention fusion for benign-malignant thyroid nodule diagnosis in ultrasound

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ARTICLE INFO

Communicated by N. Zeng

Keywords:
Thyroid nodule classification
Ultrasound imaging
Self-supervised saliency
Dual-attention fusion

ABSTRACT

Accurate benign–malignant diagnosis of thyroid nodules from ultrasound images remains challenging due to low contrast, speckle noise, and substantial inter-center variability. Most existing deep learning approaches operate directly on raw ultrasound images and rely on fully supervised learning, lacking explicit mechanisms to leverage lesion-level localization for diagnostic guidance. Although some methods incorporate segmentation to assist classification, they typically depend on fully supervised pixel-level annotations, which are often unavailable in real-world clinical settings and limit generalization to data from different clinical centers. To address these limitations, we propose a self-supervised saliency-guided dual-attention fusion framework for thyroid nodule diagnosis in ultrasound images. The proposed method learns a saliency mask to highlight diagnostically informative regions and employs self-supervised reconstruction, boundary consistency, and affinity constraints to regularize saliency learning without requiring pixel-level annotations. In addition, a lightweight dual-attention fusion module is designed to integrate original and saliency-enhanced features. Unlike conventional attention mechanisms, this module combines variance-guided channel attention with patch-wise spatial attention, enabling selective emphasis on informative channels in the original image while capturing lesion-specific contextual features guided by the saliency map. This design facilitates the learning of more discriminative representations for benign–malignant classification. Extensive experiments conducted on three multi-center thyroid ultrasound datasets, under both single-center and cross-center evaluation protocols, demonstrate that the proposed framework consistently outperforms state-of-the-art methods, with particularly notable gains in multi-center generalization performance. Code is available at https://github.com/guangguangLi/MultiModalClassifier.

1. Introduction

Thyroid nodules are among the most common endocrine disorders worldwide, with a steadily increasing incidence in recent years [1]. Although the majority of thyroid nodules are benign, malignant nodules may progress aggressively and pose serious health risks if not diagnosed and treated in a timely manner. Ultrasound imaging is the primary modality for thyroid nodule screening due to its non-invasiveness, low cost, and real-time capability [2]. However, benign–malignant diagnosis based on ultrasound remains highly dependent on radiologists' experience and is susceptible to inter-observer variability, especially in cases with subtle appearance differences. Consequently, developing reliable automated diagnostic methods is of great clinical significance, as such systems can assist radiologists, improve diagnostic consistency, and reduce clinical workload [3].

With the rapid development of deep learning, an increasing number of studies have explored computer vision techniques for automated thyroid nodule diagnosis from ultrasound images. Early works primarily adopted convolutional neural networks (CNNs), such as VGG [4], ResNet [5], DenseNet [6], and EfficientNet [7], to extract discriminative features for benign–malignant classification [8]. More recently, motivated by the strong global modeling capability of Transformer architectures, vision Transformer-based models, including ViT [9], DeiT [10], and Swin Transformer [11], have also been introduced into thyroid

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https://doi.org/10.1016/j.neucom.2026.133641
Received 27 February 2026; Received in revised form 1 April 2026; Accepted 9 April 2026
Available online 10 April 2026

ultrasound analysis to capture long-range dependencies. In addition, several task-specific designs have been proposed to address challenges inherent to ultrasound imaging, particularly the heavy speckle noise and the large variability in the shape and size of thyroid nodules. ThyFusion [12] incorporates frequency-domain mechanisms to suppress noise-related interference, while SDA-Net [13] introduces deformable attention and distillation-driven interaction aggregation to better adapt to nodules with diverse geometries.

Despite these advances, most existing methods treat the raw ultrasound image as the sole input for classification. In clinical practice, however, radiologists typically first localize the approximate lesion region and then make diagnostic decisions based on both the internal characteristics of the nodule and its surrounding context [14]. Relying solely on raw ultrasound images often leads to feature representations being contaminated by background noise and irrelevant structures, which can impair model robustness and generalization, particularly in multi-center scenarios. To alleviate this issue, several studies have explored dual-branch or multi-input diagnostic frameworks. For example, TS-DSANN [15] jointly utilizes raw images and contour information for classification, while MCDLM [16] employs a cascaded multi-task model to extract regions of interest under expert supervision and combines them with the original ultrasound images for diagnosis. A similar strategy is also adopted in TB^{2}C-Net [17], where lesion-aware information is incorporated to enhance diagnostic performance. Although dual-branch feature learning has shown promise, it remains constrained by practical limitations. As illustrated in Fig. 1, pixel-level or region-level lesion annotations are often unavailable in real-world clinical settings due to the high cost and time required for expert labeling. Moreover, even when annotation-enhanced representations are available, how to effectively leverage them to guide feature learning within and around the lesion region remains a non-trivial challenge.

To overcome these challenges, we introduce a novel framework called self-supervised saliency-guided dual-attention fusion (S^{3}DAF), which addresses several limitations of current methods. Unlike previous works, our framework eliminates the need for manually annotated lesion masks during pretraining by proposing a self-supervised approach that directly learns saliency representations from ultrasound images. This enables automatic localization of diagnostically relevant regions and distinguishes our method from others by eliminating the reliance on costly and time-consuming expert labeling, while still capturing region-specific features. Moreover, the saliency map generated by our framework is not simply an auxiliary input; it actively guides the dual-attention fusion process across multiple levels of the network. Specifically, it suppresses irrelevant background noise and enhances features corresponding to the lesion region, thus improving both the robustness and generalization of the model. This saliency-aware fusion mechanism allows for a principled integration of global lesion awareness and fine-grained image details, a unique feature not fully explored in existing literature. In contrast to traditional dual-branch methods, where additional inputs (such as segmentation masks or contours) are pre-defined or require manual intervention, our method leverages self-supervised saliency learning to adaptively discover salient regions, providing a more flexible and scalable solution.

The main contributions of this work are summarized as follows:

- We propose a self-supervised saliency-guided framework for benign–malignant thyroid nodule diagnosis that enables lesion-aware representation learning without requiring pixel-level or region-level annotations, thereby improving practicality in real clinical settings.
- We design a lightweight dual-attention fusion mechanism in which saliency-guided cues explicitly guide multi-level feature integration, suppressing noise-dominated responses in original ultrasound features while preserving discriminative lesion characteristics.
- Extensive experiments on three multi-center thyroid ultrasound datasets demonstrate that the proposed method consistently outperforms state-of-the-art approaches, exhibiting particularly strong generalization performance in cross-center evaluations.

2. Related work

2.1. Segmentation-guided disease diagnosis

Accurate identification of lesions from medical images often benefits from explicit lesion localization. Segmentation masks can serve as informative priors that guide classification networks to focus on diagnostically relevant regions, reducing the interference of background noise and improving classification performance [17,18]. MB-DCNN [19] employs a coarse segmentation network to generate lesion masks that bootstrap a mask-guided classification network, while the classification feedback further refines segmentation in a mutual learning manner. LW-XNet [20] integrates multi-level wavelet features and improved dense attention to enhance both lesion segmentation and classification. MTL-CNN [21] introduces a multi-task learning framework with edge prediction and lesion area extraction modules to exploit segmentation knowledge for more discriminative classification features. GSM [22] combines CNN-Transformer localization and U-Net refinement to fully leverage segmentation knowledge for improved tumor classification in ultrasound. MMF-U-Net [23] performs multi-modal fusion of B- and SE-mode ultrasound images for segmentation, and uses the resulting lesion masks to guide subsequent classification. In thyroid nodule diagnosis, TS-DSANN [15] employs a dual-stream attention network that leverages contour-based shape features alongside texture representations to enhance classification. MCDLM [16] utilizes a multi-task cascade model that transfers segmentation knowledge and combines it with semi-supervised image generation to assist nodule classification. These segmentation-guided approaches have demonstrated superior classification performance and advanced research in thyroid nodule diagnosis. However, in real clinical scenarios, lesion annotations are often unavailable, limiting the applicability of these methods [24]. Existing self-supervised approaches partially address this by pretraining on unlabeled data, yet most still require a small amount of labeled data for fine-tuning [25,26]. In many practical applications, acquiring even a small set of annotations can be costly or infeasible. To overcome these

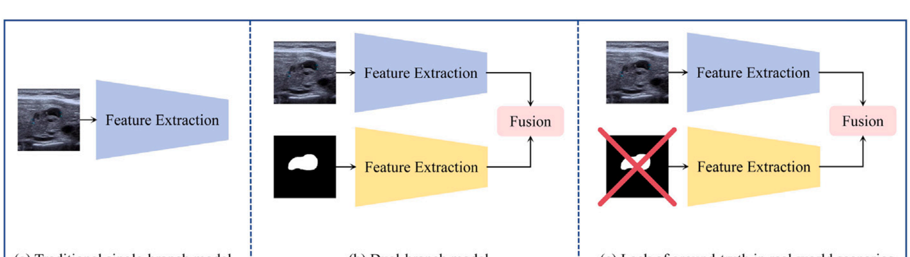


Fig. 1. (a) a single-branch model using only the original ultrasound image, (b) a dual-branch model incorporating both the image and its segmentation mask, and (c) highlights the practical limitation that ground-truth masks are often unavailable in real-world settings, hindering the deployment of dual-branch models.

limitations, we propose a novel end-to-end framework that jointly learns a saliency-guided mask and a classifier without using any manual annotations. The predicted saliency mask acts as an attention map, guiding multi-level feature fusion to improve benign–malignant classification performance.

2.2. Attention mechanisms for medical image analysis

Attention mechanisms have been widely employed in medical image analysis to enhance feature representation by selectively emphasizing diagnostically relevant information while suppressing background noise [27,28]. Among commonly used attention designs, channel attention aims to recalibrate feature responses by modeling inter-channel dependencies [29,30]. The squeeze-and-excitation (SE) [31] block adaptively reweights channels through global context aggregation, significantly boosting representational power with minimal computational overhead. To further incorporate spatial cues, CBAM [32] sequentially applies channel and spatial attention, allowing more refined feature enhancement. To reduce the information loss caused by dimensionality reduction, ECA [33] introduces efficient local cross-channel interaction without channel compression, while EMA [34] extends channel attention to a multi-level setting by grouping feature channels and integrating cross-dimension interactions. More recently, SCSA [35] investigates the synergistic relationship between spatial and channel attention, leveraging multi-semantic spatial priors to guide progressive channel-wise self-attention for stronger feature discrimination. In medical imaging applications, attention mechanisms have been extensively integrated into task-specific frameworks. For instance, MCLSF [36] exploits attention to model correlations across multiple data modalities for effective multi-client feature fusion. CA-Net [37] incorporates spatial, channel, and scale attentions simultaneously to improve segmentation accuracy and interpretability under large variations in object size and shape. Guided self-attention has also been adopted in multi-level attention (MSA) [38] frameworks to capture long-range contextual dependencies and suppress irrelevant features in medical image segmentation. Furthermore, attention has been explored in domain adaptation scenarios, such as DADASeg-Net [39], where spatial and class-wise attentions guide cross-domain alignment toward more discriminative and challenging regions. Despite their effectiveness, most existing attention mechanisms are primarily driven by feature activations from raw images and supervised task losses, lacking explicit lesion-aware guidance. Moreover, channel recalibration is often based on global average statistics, which may be insufficient for modeling subtle intra-lesion variations critical to benign–malignant discrimination in ultrasound imaging. To address these limitations, we propose a task-oriented attention mechanism that combines variance-guided channel attention with saliency-guided patch-wise spatial attention, enabling more discriminative and generalizable feature representations for thyroid nodule diagnosis.

3. Method

3.1. Overall architecture

To address the challenges of low contrast, speckle noise, and inter-center variability in thyroid ultrasound images, we propose a self-supervised saliency-guided dual-attention fusion framework for benign–malignant classification. As illustrated in Fig. 2, the framework consists of two tightly coupled components: a segmentation-based saliency generation module and a saliency-enhanced classification network, which are trained jointly in an end-to-end manner.

Given an input ultrasound image, the segmentation branch first predicts a probability-based saliency map without requiring pixel-level annotations. This map aims to highlight diagnostically relevant regions, such as lesion boundaries and internal texture patterns. To regularize saliency learning, three self-supervised losses are introduced: a reconstruction loss, a boundary consistency loss, and an affinity loss. These constraints encourage the predicted mask to preserve structural information, align with anatomical edges, and emphasize local gradient variations, thereby improving robustness to cross-center domain shifts.

The predicted saliency map is then applied to the original image via element-wise multiplication, producing a saliency-enhanced image

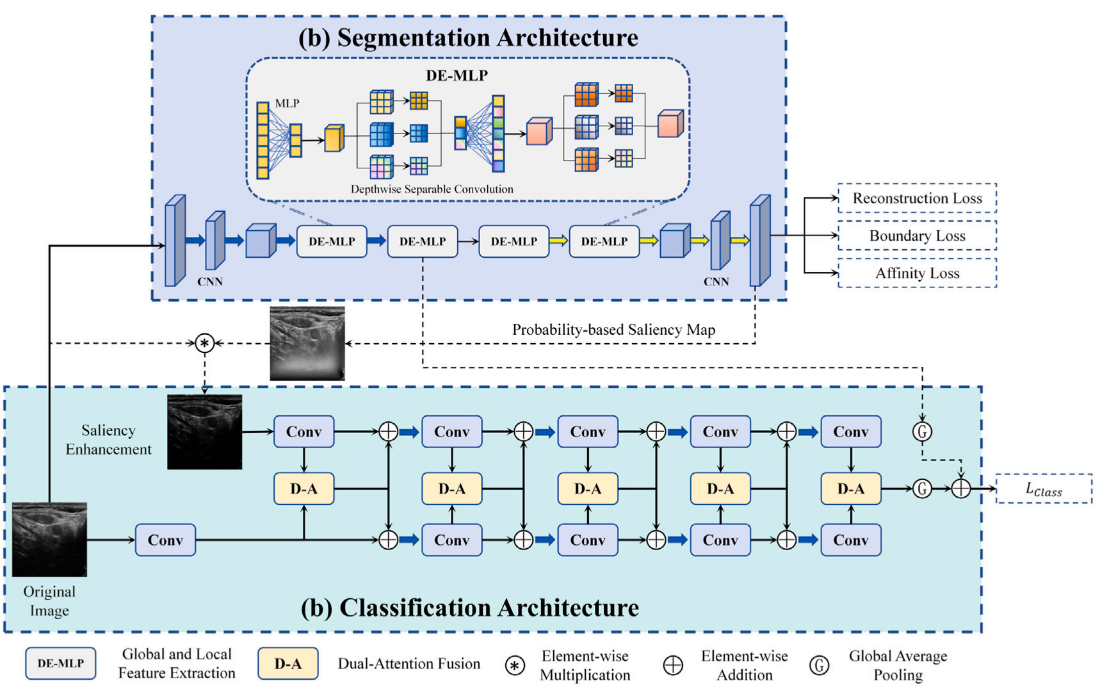


Fig. 2. A U-shaped segmentation branch, with channel dimensions [8, 16, 24, 32, 48, 64], generates a saliency map that enhances the input image via element-wise multiplication. The original and enhanced images are subsequently processed by a five-stage multi-scale dual-branch encoder (with channels increasing from 32 to 512), where the two branches do not share weights. At each stage, feature representations are integrated using a dual-attention fusion module. The resulting features, combined with a projected segmentation embedding, are finally fed into a classifier to produce the diagnostic prediction.

in which informative regions are amplified while background interference is suppressed. The original and enhanced images are fed into a dual-stream structure, where each stream has its own encoder that independently learns features from its respective input image.

At each level, a Dual-Attention (DA) fusion module integrates features from the two pathways. Specifically, variance-guided channel attention emphasizes channels with higher discriminative variability, while patch-wise spatial attention leverages the saliency map to focus on lesion-relevant regions. This dual mechanism jointly enhances feature discriminability and spatial coherence in the fused representation.

Finally, the highest-level fused feature is globally pooled and passed through a lightweight classifier to obtain the final prediction. In addition, the final embedding from the segmentation branch is projected and fused into the classification feature, further strengthening the interaction between localization and diagnosis. The entire framework is optimized end-to-end using both self-supervised saliency losses and a supervised classification loss, enabling effective learning without dense annotations.

The overall training objective is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{class}} + \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{bound}} + \mathcal{L}_{\text{aff}}, \tag{1}$$

where $\mathcal{L}_{\text{class}}$ denotes the classification loss, and the remaining terms correspond to the self-supervised regularization losses for saliency learning.

The reconstruction loss encourages the saliency map to preserve information from the original image:

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{x\sim\mathcal{D}}\left[\|x \odot m - x\|_1\right], \tag{2}$$

where x denotes the input image and m is the predicted saliency map. Crucially, this loss alone would push the saliency map m towards an all-ones map. Its effect is balanced by the other proposed losses.

The boundary loss enforces spatial consistency between the edges in the saliency map and those in the original image:

$$\mathcal{L}_{\text{bound}} = \|\nabla x - \nabla m\|_1, \tag{3}$$

where $\nabla(\cdot)$ denotes spatial gradients computed using Sobel operators. This loss encourages the spatial patterns of gradients (both location and direction) in the saliency map m to align closely with those in the original image x, promoting sharp and accurately located boundaries.

The affinity loss complements the boundary loss by comparing the local strengths of structural transitions:

$$\mathcal{L}_{\text{aff}} = \||\nabla x| - |\nabla m|\|_2^2, \tag{4}$$

where $|\cdot|$ represents the magnitude of the gradient vector. This loss focuses on ensuring that the intensity of structural features in the original image is reflected in the corresponding regions of the saliency map. By operating on gradient magnitudes rather than raw vectors, it provides a more robust measure of feature importance that is less sensitive to minor spatial misalignments. Together, these losses ensure that the saliency map not only has correctly positioned and shaped boundaries ($\mathcal{L}_{\text{bound}}$) but also accurately reflects the relative strength of features across different regions ($\mathcal{L}_{\text{aff}}$).

These three self-supervised losses work in conjunction: while the reconstruction loss prevents the suppression of all information, the boundary and affinity losses ensure that the learned saliency map remains spatially structured and highlights intrinsic features of the image, preventing it from collapsing into a trivial, constant value.

The classification branch is optimized using the standard cross-entropy loss:

$$\mathcal{L}_{\text{class}} = -\sum_{i=1}^{N} y_i \log(p_i), \tag{5}$$

where y_i and p_i denote the ground-truth label and predicted probability for the i-th sample, respectively.

3.2. Segmentation architecture

As shown in Fig. 2(b), the segmentation branch is designed to generate a probability-based saliency map highlighting diagnostically relevant regions in thyroid ultrasound images. It employs an enhanced U-shaped encoder–decoder architecture, where both encoder and decoder stages are composed of DE-MLP blocks. This design enables the effective integration of global contextual dependencies and local spatial structures, which is crucial for ultrasound images characterized by speckle noise and ambiguous boundaries.

A DE-MLP block operates on an input feature $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$ by first flattening the spatial dimensions:

$$\mathbf{x}_{\text{flat}} \in \mathbb{R}^{B\times N\times C}, \quad N = H \times W. \tag{6}$$

It then applies two stages of linear projection followed by depthwise convolution:

$$\mathbf{z}_1 = \text{DWConv}_1(\text{Linear}_1(\mathbf{x}_{\text{flat}})), \tag{7}$$

$$\mathbf{z}_2 = \text{DWConv}_2(\text{Linear}_2(\mathbf{z}_1)), \tag{8}$$

where the linear projections capture global dependencies, and the depthwise convolutions model local spatial coherence. The output is subsequently reshaped back to the original spatial dimensions for decoding.

By combining MLP-style global receptive fields with convolutional inductive biases, the DE-MLP block overcomes the limitations of conventional CNNs with restricted receptive fields and pure MLP-based models that lack spatial awareness [40]. This hybrid design is particularly effective for capturing subtle morphological features in thyroid nodules, such as irregular margins, microcalcifications, and heterogeneous echotexture.

At the output stage, a 1×1 convolution produces a logits map $\mathbf{m}_{\text{logits}} \in \mathbb{R}^{B\times1\times H\times W}$, which is converted into a probability-based saliency map via a sigmoid function:

$$\mathbf{m} = \sigma(\mathbf{m}_{\text{logits}}), \tag{9}$$

where $\sigma(\cdot)$ denotes the sigmoid activation. The resulting saliency map serves as a soft attention mask, emphasizing lesion-related regions while suppressing background noise.

Additionally, a global average pooling operation is applied to the deepest encoder feature to obtain a compact segmentation embedding $\mathbf{f}_{\text{seg}} \in \mathbb{R}^{C_5}$. This embedding is later fused with classification features, reinforcing the consistency between lesion localization and diagnostic prediction.

3.3. Classification architecture

The classification branch, illustrated in Fig. 2(b), leverages both the original ultrasound image and the saliency-enhanced image generated by the segmentation branch. These two streams are processed in parallel through a dual encoder, with feature integration at each level performed by the proposed Dual-Attention (DA) fusion module. This design allows the model to exploit complementary information: the original image provides full context, while the saliency-enhanced image emphasizes diagnostically relevant regions.

Given an input image $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$, the saliency-enhanced version is computed as

$$\mathbf{x}_{\text{enh}} = \mathbf{x} \odot \mathbf{m}, \tag{10}$$

where $\mathbf{m} \in [0,1]^{B\times1\times H\times W}$ is the sigmoid-transformed saliency map from the segmentation branch.

Both $\mathbf{x}$ and $\mathbf{x}_{\text{enh}}$ are then fed into the dual encoder. At each level, features from the two streams are fused using the DA module, producing a hierarchical representation that captures both fine-grained textures and global structures.

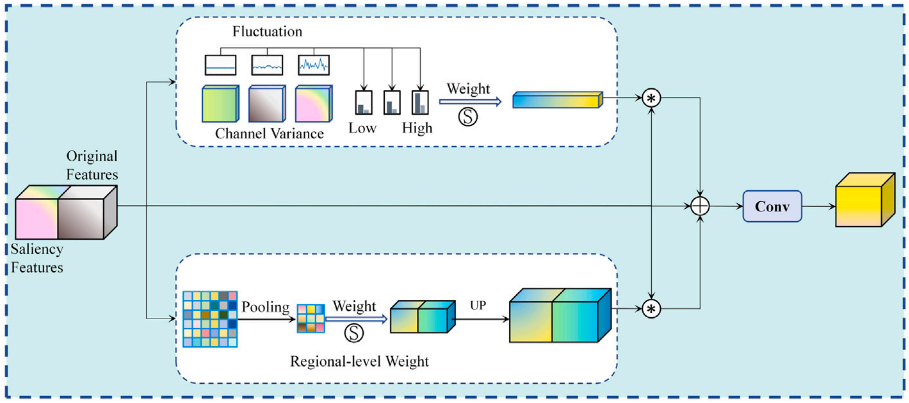


Fig. 3. The original features are processed by two parallel branches: a channel attention branch that computes spatial variance to generate channel weights, and a spatial attention branch that applies average pooling and sigmoid-weighted upsampling to produce region-level spatial weights. The two attention-weighted features are combined by element-wise summation and refined through a convolutional layer to obtain the fused output.

The DA module is the core innovation of the classification branch (Fig. 3). It combines variance-guided channel attention and patch-wise spatial attention to dynamically reweight features based on diagnostic relevance. Specifically, given feature maps $\mathbf{x}_1$ and $\mathbf{x}_2$ from the original and enhanced streams, respectively, they are concatenated along the channel dimension:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] \in \mathbb{R}^{B\times 2C\times H\times W}. \tag{11}$$

Channel attention aims to emphasize feature channels with high spatial variability, which often correlate with diagnostically relevant patterns in ultrasound imaging. We compute the spatial variance of each channel as:

$$\mathbf{v} = \mathrm{Var}_{(H,W)}(\mathbf{x}), \tag{12}$$

$$\mathbf{w}_c = \sigma(\mathbf{v}), \tag{13}$$

$$\mathbf{x}_c = \mathbf{x} \odot \mathbf{w}_c, \tag{14}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The rationale for using variance lies in its ability to capture second-order statistics of feature distributions. Compared with first-order measures such as global average pooling, variance better reflects spatial heterogeneity. In ultrasound images, lesion regions and structural distortions typically induce stronger spatial fluctuations, while background regions remain relatively homogeneous. Therefore, channels with higher variance are more likely to encode discriminative information.

Spatial attention focuses on localizing lesion-relevant regions via coarse-grained spatial modeling. Instead of pixel-wise attention, we employ a patch-wise pooling strategy:

$$\mathbf{p} = \mathrm{AvgPool}(\mathbf{x}, \mathrm{kernel\_size} = 2), \tag{15}$$

$$\mathbf{a}_s = \sigma(\mathbf{p}), \tag{16}$$

$$\mathbf{a}_s^{\uparrow} = \mathrm{Interpolate}(\mathbf{a}_s, \mathrm{size} = (H, W)), \tag{17}$$

$$\mathbf{x}_s = \mathbf{x} \odot \mathbf{a}_s^{\uparrow}. \tag{18}$$

Patch-wise pooling aggregates features within local neighborhoods, capturing mesoscopic structural patterns while suppressing high-frequency speckle noise. This yields smoother and more robust attention maps that are less sensitive to local misalignments. Compared to pixel-wise mechanisms such as CBAM, the proposed approach introduces implicit spatial regularization and better preserves lesion-level consistency.

The channel- and spatial-attentive features are combined and projected back to the original channel dimension:

$$\mathbf{x}_{\mathrm{fused}} = \mathbf{x}_c + \mathbf{x}_s, \tag{19}$$

$$\mathbf{y} = \mathrm{Conv}_{1\times 1}(\mathbf{x}_{\mathrm{fused}}) \in \mathbb{R}^{B\times C\times H\times W}. \tag{20}$$

This dual-pathway attention effectively balances *global statistical significance* and *local anatomical consistency*, making the representation robust to low contrast, speckle noise, and ambiguous boundaries.

Finally, the highest-level feature map $\mathbf{F}_{\mathrm{last}}$ is globally pooled:

$$\mathbf{f}_{\mathrm{cls}} = \mathrm{GlobalAvgPool}(\mathbf{F}_{\mathrm{last}}), \tag{21}$$

and combined with the segmentation embedding $\mathbf{f}_{\mathrm{seg}}$ via a projection layer:

$$\mathbf{f}_{\mathrm{final}} = \mathbf{f}_{\mathrm{cls}} + \mathrm{Proj}(\mathbf{f}_{\mathrm{seg}}), \tag{22}$$

reinforcing the link between lesion localization and diagnostic prediction. The effectiveness of this dual-attention design is further validated in our ablation studies.

4. Experiments

4.1. Implementation details and datasets

All experiments were conducted on a single NVIDIA RTX 8000 GPU under a unified training environment to ensure experimental consistency. The proposed network was trained for 150 epochs with a batch size of 64 using the Adam optimizer. The initial learning rate was set to 1×10^{-4} and kept constant throughout training. To ensure reproducibility, all random seeds were fixed prior to model initialization and data loading. Ultrasound images were categorized into two diagnostic classes, namely *benign* and *malignant*. Before training, all images were resized to 224×224 pixels and converted to three-channel RGB format. For each dataset, the samples were randomly divided into training and testing subsets with an 80%/20% split.

This study was conducted in accordance with institutional ethical standards and was approved by the Research Ethics Committees of the Sun Yat-sen University Cancer Center and the Hospital of Northwestern Polytechnical University. All ultrasound images were retrospectively collected, fully anonymized prior to analysis, and informed consent was waived in compliance with relevant regulations.

The proposed method was evaluated on a diverse collection of thyroid nodule ultrasound datasets, including both a large-scale public benchmark and multiple private multi-center cohorts. Specifically, the experimental evaluation involved the following datasets:

- **TN5000** [41]: A large publicly available thyroid ultrasound dataset consisting of 5000 images, including 3572 malignant and 1428 benign cases.

- **TN-SYUCC**: A private thyroid nodule dataset collected at the Sun Yat-sen University Cancer Center, comprising 1167 ultrasound images (648 malignant and 519 benign).
- **TN-HNPU**: A private dataset acquired from the Hospital of Northwestern Polytechnical University, including 285 thyroid nodule images with a balanced distribution of benign and malignant cases (141 malignant, 144 benign).

Together, these datasets span multiple clinical centers, imaging devices, and anatomical targets, providing a comprehensive and challenging benchmark for evaluating both in-domain performance and cross-center generalization capabilities of the proposed framework.

4.2. Comparison with state-of-the-art (SOTA) methods

To comprehensively evaluate the effectiveness of the proposed S^3DAF framework, we compared it with several representative CNN- and Transformer/Mamba-based models. The evaluation was conducted on three multi-center thyroid ultrasound datasets (TN5000, TN-SYUCC, and TN-HNPU), under both single-center and cross-center protocols.

Table 1 reports the performance of all models on the TN5000 dataset under a single-center internal evaluation. Among the CNN-based models, DenseNet achieves the highest accuracy (80.80%) and F1 score (76.78%), with VGG-16, ResNet, EfficientNet, and GhostNet performing slightly worse in comparison. While Transformer/Mamba-based models generally underperform relative to CNNs, MedMamba achieves the highest accuracy among them, with 69.10% accuracy and 62.51% F1 score. A notable observation is that while the parameter count varies significantly across models, with DenseNet having 6.95M parameters and ResNet 11.17M parameters, Transformer/Mamba-based models such as EfficientViT are relatively more efficient, with only 2.15M parameters, albeit at the cost of performance. In contrast, our proposed S^3DAF model strikes a balance between accuracy and parameter efficiency. With just 10.26M parameters, S^3DAF not only achieves a significant performance boost (89.30% accuracy, 86.62% F1 score) but does so with fewer parameters compared to many traditional CNN-based models like VGG-16 (134.26M parameters) or DenseNet (6.95M parameters). This demonstrates that the saliency-guided dual-attention fusion mechanism in S^3DAF not only enhances discriminative feature representation in ultrasound images but does so with a relatively modest increase in computational cost.

It is important to note that while the proposed method shows substantial improvements over the baseline models, such as Transformer-based methods, several factors may contribute to this performance gap. The TN5000 dataset used in this study is not exceedingly large, which could have influenced the results. Additionally, the Transformer-based models were implemented with default hyperparameters from their original papers and were not fine-tuned for this specific dataset. Furthermore, pre-trained weights were not utilized for these models, which may have impacted their performance relative to S^3DAF, which was specifically designed and optimized for this task.

As shown in Table 2, on TN-SYUCC, DenseNet again shows strong performance among CNNs (90.17% accuracy, 89.89% F1), while Transformer-based methods such as LeViT and EfficientViT achieve moderate results (80.34–81.62% accuracy). S^3DAF surpasses all baselines with 92.73% accuracy and 92.54% F1 score. This demonstrates that the proposed framework can robustly capture lesion-specific information even when image characteristics differ from TN5000, benefiting from the self-supervised saliency learning.

Table 3 presents the performance on TN-HNPU. CNN-based DenseNet achieves the best baseline result (94.73% accuracy, 94.67% F1 score), while Transformer/Mamba-based models lag behind, with MedMamba reaching 85.96% accuracy and 85.85% F1. Our method attains 96.49%

Table 1
Comparison of the proposed S^3DAF method with CNN- and Transformer/Mamba-based models on the TN5000 dataset.

Type	TN5000					
	Networks ↑	Accuracy (%) ↑	F1 score (%) ↑	Precision (%) ↑	Recall (%) ↑	Param (M) ↓
CNN-based	VGG-16 [4]	78.30	74.70	74.07	75.60	134.26
	ResNet [5]	72.40	63.86	65.82	63.08	11.17
	DenseNet [6]	80.80	76.78	76.89	76.67	6.95
	EfficientNet [7]	69.10	60.38	61.42	59.95	4.01
	GhostNet [42]	63.30	57.80	57.64	58.32	3.90
Transformer/Mamba-based	DeiT [10]	68.70	59.87	60.87	59.47	85.64
	PiT [43]	66.50	56.41	57.41	56.22	72.49
	LeViT [44]	69.80	57.75	61.04	57.57	7.00
	EfficientViT [45]	63.50	56.91	56.81	57.07	**2.15**
	CrossFormer [46]	66.40	60.58	60.38	60.92	27.51
	MedMamba [47]	69.10	62.51	62.61	62.43	13.25
	S^3DAF (Ours)	**89.30**	**86.62**	**88.25**	**85.37**	10.26

Table 2
Comparison of the proposed S^3DAF method with CNN- and Transformer/Mamba-based models on the TN-SYUCC dataset.

Type	TN-SYUCC				
	Networks ↑	Accuracy (%) ↑	F1 score (%) ↑	Precision (%) ↑	Recall (%) ↑
CNN-based	VGG-16 [4]	82.47	81.86	82.03	81.72
	ResNet [5]	83.76	83.22	83.33	83.12
	DenseNet [6]	90.17	89.89	89.83	89.95
	EfficientNet [7]	82.47	82.38	82.71	83.68
	GhostNet [42]	79.48	78.50	79.21	78.11
Transformer/Mamba-based	DeiT [10]	75.21	74.61	74.50	74.76
	PiT [43]	74.35	73.93	73.80	74.33
	LeViT [44]	80.34	80.13	80.08	80.95
	EfficientViT [45]	81.62	81.20	81.03	81.44
	CrossFormer [46]	77.77	76.96	77.16	76.80
	MedMamba [47]	75.64	74.32	75.23	73.92
	S^3DAF (Ours)	**92.73**	**92.54**	**92.38**	**92.74**

Table 3
Comparison of the proposed S^3DAF method with CNN- and Transformer/Mamba-based models on the TN-HNPU dataset.

Type	TN-HNPU				
	Networks ↑	Accuracy (%) ↑	F1 score (%) ↑	Precision (%) ↑	Recall (%) ↑
CNN-based	VGG-16 [4]	80.70	80.48	80.37	81.06
	ResNet [5]	91.22	91.05	90.87	91.28
	DenseNet [6]	94.73	94.67	94.44	95.45
	EfficientNet [7]	82.45	81.77	82.33	81.43
	GhostNet [42]	84.21	83.70	83.95	83.52
Transformer/Mamba-based	DeiT [10]	82.45	82.00	82.00	82.00
	PiT [43]	78.94	78.13	78.63	77.84
	LeViT [44]	77.19	76.46	76.66	76.32
	EfficientViT [45]	71.92	71.85	72.40	72.91
	CrossFormer [46]	80.70	80.08	80.30	79.92
	MedMamba [47]	85.96	85.85	85.83	86.74
	S^3DAF (Ours)	**96.49**	**96.43**	**96.15**	**96.96**

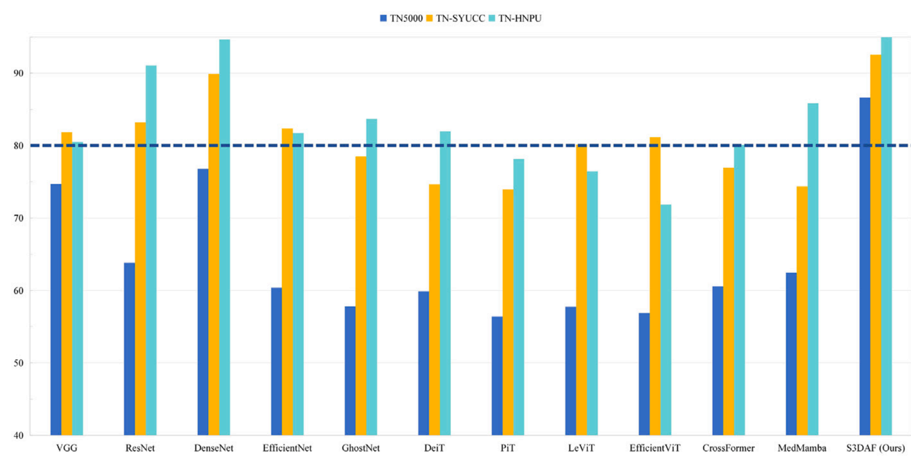


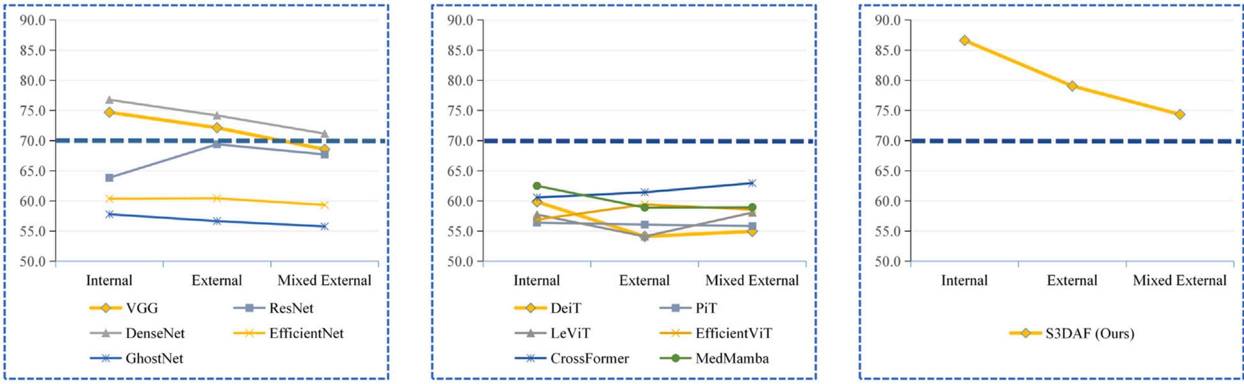
Fig. 4. Comparison of F1 scores across three single-center datasets.

Table 4
Performance comparison of different models on the TN5000 (training and internal testing), TN-SYUCC (external testing), and mixed external testing datasets (TN-SYUCC + TN-HNPU).

Type	Networks	Internal		External		Mixed External	
		Acc (%) ↑	F1 (%) ↑	Acc (%) ↑	F1 (%) ↑	Acc (%) ↑	F1 (%) ↑
CNN-based	VGG-16 [4]	78.30	74.70	73.26	72.15	69.55	68.56
	ResNet [5]	72.40	63.86	70.52	69.40	68.52	67.72
	DenseNet [6]	80.80	76.78	74.20	74.18	72.17	71.17
	EfficientNet [7]	69.10	60.38	63.92	60.44	61.70	59.34
	GhostNet [42]	63.30	57.80	60.58	56.66	56.68	55.78
Transformer/Mamba-based	DeiT [10]	68.70	59.87	56.72	54.12	57.36	54.97
	PiT [43]	66.50	56.41	60.49	56.08	58.88	55.84
	LeViT [44]	69.80	57.75	56.72	54.12	58.81	58.08
	EfficientViT [45]	63.50	56.91	60.23	59.42	60.12	58.57
	CrossFormer [46]	66.40	60.58	64.95	61.44	65.08	62.96
	MedMamba [47]	69.10	62.51	63.15	58.91	62.39	58.96
	S^3DAF (Ours)	**89.30**	**86.62**	**79.26**	**79.06**	**74.58**	**74.35**

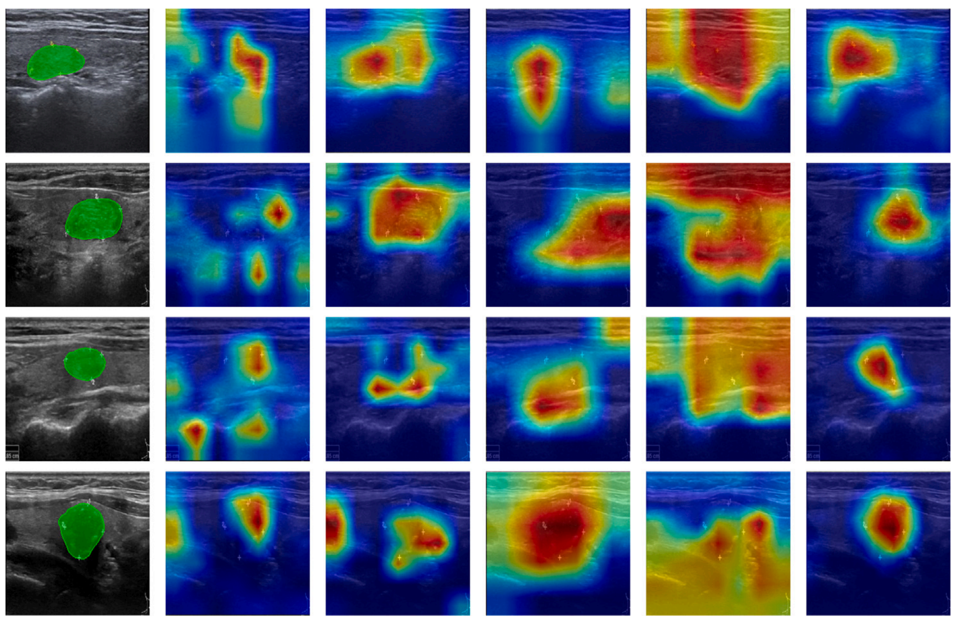
accuracy and 96.43% F1 score, confirming that S^3DAF consistently enhances classification performance across different centers by emphasizing diagnostically relevant regions and integrating multi-level context. Fig. 4 visually compares the performance of our S^3DAF method with other approaches across three datasets. Notably, S^3DAF is the only method that consistently achieves an F1 score above 80% on all datasets. In contrast, most baseline methods exhibit noticeable performance fluctuations across different datasets, with several approaches failing to reach the 80% threshold in at least one setting. Even methods that achieve relatively strong results on individual datasets (e.g., DenseNet or MedMamba) do not maintain consistent performance across all centers, indicating limited robustness to inter-center variability. This indicates that, when trained on a single-center dataset, our method can effectively capture and leverage center-specific characteristics, leading to strong generalization performance.

To evaluate generalization ability, we trained models on TN5000 and performed external testing on TN-SYUCC as well as mixed external testing (TN-SYUCC + TN-HNPU), summarized in Table 4. CNN-



(a) CNN-based (b) Transformer/Mamba-based (c) S³DAF (Ours)

Fig. 5. F1 scores of various models under internal, external (TN-SYUCC), and mixed external (TN-SYUCC + TN-HNPU) evaluation settings.



(a) Images+GT (b) VGG (c) DenseNet (d) EfficientViT (e) CrossFormer (f) Ours

Fig. 6. Activation maps on the TN5000 dataset. The ground-truth masks were manually annotated by our clinical team for qualitative comparison only and were not used during training, as the original dataset does not provide pixel-level labels.

and Transformer-based models show noticeable performance drops under external evaluation, reflecting limited cross-center robustness. For instance, DenseNet drops from 80.80% internal accuracy to 74.20% on TN-SYUCC and 72.17% on mixed external testing. In contrast, S^{3}DAF maintains high performance with 79.26% accuracy on TN-SYUCC and 74.58% on mixed external datasets, demonstrating superior cross-center generalization. This highlights the advantage of self-supervised saliency regularization and dual-attention fusion in handling variations in imaging protocols and nodule appearance. Fig. 5 illustrates line charts of F1 scores under internal, external (TN-SYUCC), and mixed external (TN-SYUCC + TN-HNPU) evaluation settings. CNN-based models (a) exhibit pronounced performance degradation under domain shift; for example, DenseNet decreases from approximately 76.0 to 70.0, and similar downward trends can be observed for VGG and EfficientNet, indicating limited cross-center robustness. Transformer- and Mamba-based models (b) generally underperform CNNs in internal evaluations and experience further deterioration in external settings, with several models dropping below 60% on the mixed dataset, reflecting their sensitivity to distribution shifts and insufficient generalization capability. In contrast, our S^{3}DAF (c) maintains consistently high and stable F1 scores across all settings, with only marginal performance degradation from internal to mixed external evaluation. Notably, its performance remains well above the 70% reference line in all scenarios, demonstrating strong resilience to domain shifts. This stability suggests that S^{3}DAF is able to learn more transferable and domain-invariant representations, thereby effectively mitigating the adverse impact of inter-center variability. Such robustness is critical for real-world deployment, where training and testing data are often collected from heterogeneous sources.

This improved generalization can be attributed to the proposed saliency learning strategy. By explicitly guiding the model to focus on lesion-relevant regions while suppressing background noise, the learned representations become less dependent on center-specific imaging characteristics. In particular, the saliency-driven supervision encourages the network to capture structural and morphological cues that are intrinsically associated with the pathology, rather than superficial appearance variations. As a result, the model reduces its reliance on domain-specific statistics and learns more invariant features across different data distributions. This mechanism helps explain why S^{3}DAF maintains relatively stable performance on external datasets despite noticeable domain shifts.

Table 5
Impact of Self-Supervised Loss Functions on accuracy and F1 score.

Reconstruction Loss	Boundary Loss	Affinity Loss	Accuracy (%)	F1 Score (%)
✗	✗	✗	84.50	81.83
✓	✗	✗	87.10	83.95
✗	✓	✗	87.30	84.89
✗	✗	✓	86.60	83.04
✓	✓	✓	**89.30**	**86.62**

Table 6
The performance of the dual-attention mechanism is compared with existing attention mechanisms under the incorporation of a self-supervised loss function.

Attention Mechanisms	Accuracy (%)	F1 Score (%)
+ SE [31]	86.90	83.80
+ CBAM [32]	82.10	78.90
+ EMA [33]	85.20	81.09
+ ECA [34]	87.70	84.41
+ SCSA [35]	87.10	84.15
+ Dual-Attention (ours)	**89.30**	**86.62**

As shown in the Fig. 6, the activation map comparison clearly highlights the advantages of S^3DAF (Ours) in thyroid nodule localization and attention modeling. Compared with existing models such as VGG, DenseNet, EfficientViT, and CrossFormer, which often exhibit dispersed activations, blurred boundaries, or spurious responses to background regions, the proposed self-supervised saliency-guided dual-attention fusion framework accurately concentrates on the true nodule regions. The resulting activation heatmaps are highly focused, with contours that closely align with the ground-truth annotations. Moreover, even under challenging conditions–including low contrast and severe speckle noise–our method consistently emphasizes clinically relevant diagnostic regions while effectively suppressing irrelevant background interference. These results demonstrate that S^3DAF achieves superior sensitivity, robustness, and interpretability, and importantly, such performance is obtained without requiring pixel-level annotations during training, highlighting its practical value for real-world clinical deployment where fine-grained annotations are often difficult to obtain.

4.3. Ablation study

To systematically analyze the contribution of each key component in the proposed S^3DAF framework, we conducted a series of ablation experiments on the TN5000 dataset. Specifically, we investigate: (1) the impact of different self-supervised saliency regularization losses, including reconstruction loss, boundary consistency loss, and affinity loss, on classification performance; and (2) the effectiveness of the proposed dual-attention fusion module compared with existing attention mechanisms commonly adopted in medical image analysis. All ablation experiments are performed under the same experimental settings to ensure a fair comparison. Table 5 reports the impact of different self-supervised loss combinations on classification accuracy and F1 score. When none of the self-supervised saliency constraints are applied, the model is trained solely with the classification loss, resulting in relatively limited performance (84.50% accuracy and 81.83% F1 score). Introducing individual self-supervised losses consistently improves performance. Specifically, the reconstruction loss encourages the saliency mask to preserve essential structural information of the original image, leading to a notable performance gain (87.10% accuracy). The boundary consistency loss further refines lesion contours by enforcing spatial smoothness along salient region boundaries, yielding improved delineation of diagnostically relevant areas (87.30% accuracy). The affinity loss models intra-region feature consistency and inter-region discrimination, enabling more coherent saliency representations and improved F1 score. When all three loss functions are applied together, the model achieved the best performance, with an accuracy of 89.30% and an F1 score of 86.62%. These results demonstrate that the proposed self-supervised saliency learning strategy effectively provides lesion-aware guidance without requiring pixel-level annotations. The complementary nature of reconstruction, boundary, and affinity constraints ensures both structural fidelity and semantic consistency of the learned saliency maps, which in turn enhances downstream benign-malignant classification.

To further validate the design of the proposed attention mechanism, we compare the dual-attention fusion module with several widely used attention strategies, including SE [31], CBAM [32], EMA [33], ECA [34], and SCSA [35]. The results are summarized in Table 6. Conventional channel attention mechanisms such as SE and ECA moderately improve performance by reweighting global channel responses, while spatial-channel hybrid methods like CBAM and SCSA show limited gains, likely due to their reliance on generic spatial attention that is not explicitly aligned with lesion regions in ultrasound images. In contrast, the proposed dual-attention module achieves the highest performance, with 89.30% accuracy and 86.62% F1 score. This superior performance can be attributed to its tailored design for ultrasound-based diagnosis. The variance-guided channel attention effectively suppresses speckle noise and low-informative channels in the original ultrasound features, while the saliency-guided patch-wise spatial attention selectively enhances contextual representations within lesion regions highlighted by the learned saliency mask. By jointly integrating original and saliency-enhanced features, the dual-attention fusion module produces more discriminative and robust representations, thereby significantly improving classification performance.

5. Conclusion

In this study, we propose a self-supervised saliency-guided dual-attention fusion framework for benign-malignant classification of thyroid nodules from ultrasound images. Unlike conventional methods that rely solely on raw images or fully supervised segmentation, our approach leverages a learnable saliency mask regularized by reconstruction, boundary consistency, and affinity losses to emphasize diagnostically informative regions without requiring pixel-level annotations. The dual-attention fusion module effectively integrates original and saliency-enhanced features, combining variance-guided channel attention to suppress noise in the original ultrasound features with saliency-guided spatial attention to enhance lesion-specific representations, thereby producing highly discriminative features. Extensive experiments on three multi-center datasets demonstrate that our framework consistently outperforms state-of-the-art methods, achieving superior accuracy, F1 scores, and cross-center generalization, highlighting its robustness to variations in imaging protocols and clinical centers. Future work will explore adaptive attention mechanisms that dynamically adjust to varying nodule sizes and appearances, semi-supervised extensions to leverage large-scale unlabeled datasets, and integration with clinical metadata to further enhance diagnostic performance and real-world applicability.

CRediT authorship contribution statement

Xiaodu Zeng: Writing – original draft, Methodology, Investigation, Conceptualization. **Guangju Li:** Writing – original draft, Visualization, Validation, Methodology. **Jiajia Li:** Supervision. **Guanlin Li:** Supervision. **Qinghua Huang:** Writing – review & editing, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was partially supported by the Science and Technology Major Project of Guangxi (GuikeAA23073013) and the National Natural Science Foundation of China (Grants U25A20448, 12572068, and 62571414).

Data availability

Data will be made available on request.

References

[1] G. Grani, M. Sponziello, S. Filetti, C. Durante, Thyroid nodules: diagnosis and management, Nat. Rev. Endocrinol. 20 (2024) 715–728.
[2] S. Luo, E.-H. Kim, M. Dighe, Y. Kim, Thyroid nodule classification using ultrasound elastography via linear discriminant analysis, Ultrasonics 51 (2011) 425–431.
[3] J. Chi, E. Walia, P. Babyn, J. Wang, G. Groot, M. Eramian, Thyroid nodule classification in ultrasound images by fine-tuning deep convolutional neural network, J. Digit. Imaging 30 (2017) 477–486.
[4] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint arXiv:1409.1556, 2014.
[5] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
[6] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.
[7] M. Tan, Q. Le, Efficientnet: rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114.
[8] N. Aboudi, H. Khachnaoui, O. Moussa, N. Khlifa, Bilinear pooling for thyroid nodule classification in ultrasound imaging, Arab. J. Sci. Eng. 48 (2023) 10563–10573.
[9] A. Dosovitskiy, An image is worth 16x16 words: Transformers for image recognition at scale, arXiv preprint arXiv:2010.11929, 2020.
[10] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jégou, Training data-efficient image transformers & distillation through attention, in: International Conference on Machine Learning, PMLR, 2021, pp. 10347–10357.
[11] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.
[12] G. Chen, N. Zhu, J. Lin, B. Pu, H. Luo, K. Li, Thyfusion: a lightweight attribute enhancement module for thyroid nodule diagnosis using gradient and frequency-domain awareness, Neurocomputing 613 (2025) 128749.
[13] M. Li, H. Zhou, X. Li, P. Yan, Y. Jiang, H. Luo, X. Zhou, S. Yin, Sda-net: self-distillation driven deformable attentive aggregation network for thyroid nodule identification in ultrasound images, Artif. Intell. Med. 146 (2023) 102699.
[14] J. Ma, F. Wu, T. Jiang, Q. Zhao, D. Kong, Ultrasound image-based thyroid nodule automatic segmentation using convolutional neural networks, Int. J. Comput. Assist. Radiol. Surg. 12 (2017) 1895–1910.
[15] L. Tang, C. Tian, H. Yang, Z. Cui, Y. Hui, K. Xu, D. Shen, Ts-dsann: texture and shape focused dual-stream attention neural network for benign-malignant diagnosis of thyroid nodules in ultrasound images, Med. Image Anal. 89 (2023) 102905.
[16] W. Yang, Y. Dong, Q. Du, Y. Qiang, K. Wu, J. Zhao, X. Yang, M.B. Zia, Integrate domain knowledge in training multi-task cascade deep learning model for benign–malignant thyroid nodule classification on ultrasound images, Eng. Appl. Artif. Intell. 98 (2021) 104064.
[17] M. Wang, C. Chen, Z. Xu, L. Xu, W. Zhan, J. Xiao, Y. Hou, B. Huang, L. Huang, S. Li, An interpretable two-branch bi-coordinate network based on multi-grained domain knowledge for classification of thyroid nodules in ultrasound images, Med. Image Anal. 97 (2024) 103255.
[18] Y. Liu, D. Zhang, Q. Zhang, J. Han, Part-object relational visual saliency, IEEE Trans. Pattern Anal. Mach. Intell. 44 (2021) 3688–3704.
[19] Y. Xie, J. Zhang, Y. Xia, C. Shen, A mutual bootstrapping model for automated skin lesion segmentation and classification, IEEE Trans. Med. Imaging 39 (2020) 2482–2493.
[20] X. Zheng, Y. Huang, W. Liu, C. Cai, Lw-xnet for segmentation and classification of skin lesions from dermoscopy images, Expert Syst. Appl. 255 (2024) 124826.
[21] X. He, Y. Wang, S. Zhao, X. Chen, Joint segmentation and classification of skin lesions via a multi-task learning convolutional neural network, Expert Syst. Appl. 230 (2023) 120174.
[22] T. Jiang, J. Guo, W. Xing, M. Yu, Y. Li, B. Zhang, Y. Dong, D. Ta, A prior segmentation knowledge enhanced deep learning system for the classification of tumors in ultrasound image, Eng. Appl. Artif. Intell. 142 (2025) 109926.
[23] Y. Cho, S. Misra, R. Managuli, R.G. Barr, J. Lee, C. Kim, Attention-based fusion network for breast cancer segmentation and classification using multi-modal ultrasound images, Ultrasound Med. Biol. 51 (2025) 568–577.
[24] A. Kalapos, B. Gyires-Tóth, Self-supervised pretraining for 2d medical image segmentation, in: European Conference on Computer Vision, Springer, 2022, pp. 472–484.
[25] C. Ouyang, C. Biffi, C. Chen, T. Kart, H. Qiu, D. Rueckert, Self-supervised learning for few-shot medical image segmentation, IEEE Trans. Med. Imaging 41 (2022) 1837–1848.
[26] C. Chen, J. Han, K. Debattista, Virtual category learning: a semi-supervised learning method for dense prediction with extremely limited labels, IEEE Trans. Pattern Anal. Mach. Intell. 46 (2024) 5595–5611.
[27] X. Li, M. Li, P. Yan, G. Li, Y. Jiang, H. Luo, S. Yin, Deep learning attention mechanism in medical image analysis: basics and beyonds, Int. J. Netw. Dyn. Intell. (2023) 93–116.
[28] X. Li, W. Tan, X. Huang, Y. Zhang, H. Zeng, T. Chen, N. Zeng, Paws-net: prior-aware weakly supervised network for pulmonary disease localization and classification in CT scans, Neurocomputing (2026) 132811.
[29] L. Hu, Z. Wang, P. Wu, K. Yu, N. Zeng, DartsNeXt: bridging the search-evaluation gap in differentiable NAS with router-based selection and sequential architecture, IEEE Trans. Emerg. Top. Comput. Intell. (2026) 1–13, https://doi.org/10.1109/TETCI.2026.3670670.
[30] Y. Liu, C. Li, S. Xu, J. Han, Part-whole relational fusion towards multi-modal scene understanding, Int. J. Comput. Vis. 133 (2025) 4483–4503.
[31] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.
[32] S. Woo, J. Park, J.-Y. Lee, I.S. Kweon, CBAM: convolutional block attention module, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 3–19.
[33] T. Liu, R. Luo, L. Xu, D. Feng, L. Cao, S. Liu, J. Guo, Spatial channel attention for deep convolutional neural networks, Mathematics 10 (2022) 1750.
[34] D. Ouyang, S. He, G. Zhang, M. Luo, H. Guo, J. Zhan, Z. Huang, Efficient multi-scale attention module with cross-spatial learning, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2023, pp. 1–5.
[35] Y. Si, H. Xu, X. Zhu, W. Zhang, Y. Dong, Y. Chen, H. Li, Scsa: exploring the synergistic effects between spatial and channel attention, Neurocomputing 634 (2025) 129866.
[36] M. Irfan, K.M. Malik, K. Muhammad, Federated fusion learning with attention mechanism for multi-client medical image analysis, Inf. Fusion 108 (2024) 102364.
[37] R. Gu, G. Wang, T. Song, R. Huang, M. Aertsen, J. Deprest, S. Ourselin, T. Vercauteren, S. Zhang, Ca-net: comprehensive attention convolutional neural networks for explainable medical image segmentation, IEEE Trans. Med. Imaging 40 (2020) 699–711.
[38] A. Sinha, J. Dolz, Multi-scale self-guided attention for medical image segmentation, IEEE J. Biomed. Health Inform. 25 (2020) 121–130.
[39] X. Chen, T. Kuang, H. Deng, S.H. Fung, J. Gateno, J.J. Xia, P.-T. Yap, Dual adversarial attention mechanism for unsupervised domain adaptive medical image segmentation, IEEE Trans. Med. Imaging 41 (2022) 3445–3453.
[40] J.M.J. Valanarasu, V.M. Patel, Unext: mlp-based rapid medical image segmentation network, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2022, pp. 23–33.
[41] H. Zhang, Q. Liu, X. Han, L. Niu, W. Sun, Tn5000: an ultrasound image dataset for thyroid nodule detection and classification, Sci. Data 12 (2025) 1437.
[42] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, C. Xu, Ghostnet: more features from cheap operations, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 1580–1589.
[43] B. Heo, S. Yun, D. Han, S. Chun, J. Choe, S.J. Oh, Rethinking spatial dimensions of vision transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 11936–11945.
[44] B. Graham, A. El-Nouby, H. Touvron, P. Stock, A. Joulin, H. Jégou, M. Douze, Levit: a vision transformer in convnet's clothing for faster inference, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 12259–12269.
[45] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, Y. Yuan, Efficientvit: memory efficient vision transformer with cascaded group attention, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 14420–14430.
[46] W. Wang, W. Chen, Q. Qiu, L. Chen, B. Wu, B. Lin, X. He, W. Liu, Crossformer++: a versatile vision transformer hinging on cross-scale attention, IEEE Trans. Pattern Anal. Mach. Intell. 46 (2023) 3123–3136.
[47] Y. Yue, Z. Li, Medmamba: Vision mamba for medical image classification, arXiv preprint arXiv:2403.03849, 2024.

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